

Portfolio

CutMix-SupCon: Supervised Contrastive Learning with CutMix Augmentation for Multimodal-based Emotion Recognition

서론

- 멀티모달 데이터는 구조적으로 획득하기 어려운 문제가 있으며, 부족한 데이터 샘플의 문제로 인해 학습된 딥러닝 기반 네트워크의 실용화가 진행되기에는 여전히 한계점이 존재한다.
- 선행 연구들은 실제 환경의 부족한 데이터 문제는 고려하지 않고, 멀티모달 결합 관점에 대해서만 집중되는 경향이 존재한다.
- 본 연구에서는 멀티모달 데이터셋의 불충분한 문제를 해결하기 위해 CutMix-SupCon 방법을 제안한다. 제안된 CutMix-SupCon 방법은 잠재 공간에서 추출된 음성-텍스트 모달 간의 결합된 특징표현으로부터 혼합된 특징표현을 만들고, 혼합 대조 손실을 통해 음성-텍스트의 결합된 멀티모달 특징 표현이 원본과 얼마나 유사한지에 대한 목적 함수를 통해 최적화하였다.

방법

제안하는 멀티모달 CutMix-SupCon 방법은

- 멀티모달 특징표현 결합
 - CutMix 기반 멀티모달 혼합 증강
 - SupCon
- 으로 나뉘어지며, 전체 흐름도는 다음과 같다.

- CutMix 기반 멀티모달 혼합 증강
멀티모달 특징표현 z 와 레이블 y 에 대하여 샘플 $(z_A, y_A), (z_B, y_B)$ 를 구성할 수 있고, 이로부터 새로운 샘플 $(\tilde{z}, \tilde{y})$ 를 아래의 식과 같이 만들 수 있다.
$$\tilde{z} = M \odot z_A + (1 - M) \odot z_B$$
$$\tilde{y} = \lambda y_A + (1 - \lambda) y_B$$

- CutMix-SupCon
CutMix 기법으로부터 증강된 멀티모달 특징표현을 위한 적절한 손실 함수를 적용하기 위해, 잠재 공간내에서 혼합된 정보를 구분하여 효과적인 학습을 가능하게 하는 CutMix-SupCon 방법을 제안한다.

$$L_{CL} = \sum_{i \in I} \frac{1}{|P(i)|} \sum_{p \in P(i)} \log \frac{\exp(z_i \cdot \frac{\tilde{z}_p}{\tau})}{\sum_{a \in A(i)} \exp(z_i \cdot \frac{\tilde{z}_a}{\tau})}$$
$$L = L_{CE} + \alpha L_{CL}$$

위와 같은 최종 손실 함수를 통해 모델은 혼합된 멀티모달 특징표현이 원본의 샘플의 고유 정보를 얼마나 포함하는지를 학습한다.

실험 및 결과

- 데이터셋: KEMDy19 Text+Audio 멀티모달 감정인식 데이터셋
중복된 파일, 음성의 발화 세그먼트, 20초 이상의 음성 파일 제거 등의 전처리를 통해 10,214개의 데이터 샘플을 확보
학습 및 평가셋 8:2 무작위 분할
- 방법: 각 모달에 대한 특징표현 기반의 CE, Mixup, CutMix, 제안하는 Cutmix-SupCon 방법에 대한 결과 보고

Input	Method	Acc (%)	F1-score (%)
Text	Cross-Entropy	73.42	69.32
Audio		72.59	67.13
Text+Audio		73.57	69.08
Text	Mixup	71.76	67.37
Audio		71.46	65.89
Text+Audio		72.39	68.94
Text	CutMix	72.93	69.37
Audio		71.76	65.94
Text+Audio		74.74	70.77
Text	CutMix-SupCon (ours)	72.49	69.72
Audio		72.54	68.27
Text+Audio		75.28	71.17

표 1 KEMDy19 데이터셋을 사용한 감정인식 평가 결과

Input	Method	Weighted F1-score(%)
Text+Audio	Cross-Entropy	73.55
	Mixup	72.39
	CutMix	74.77
	CutMix-SupCon (ours)	74.96

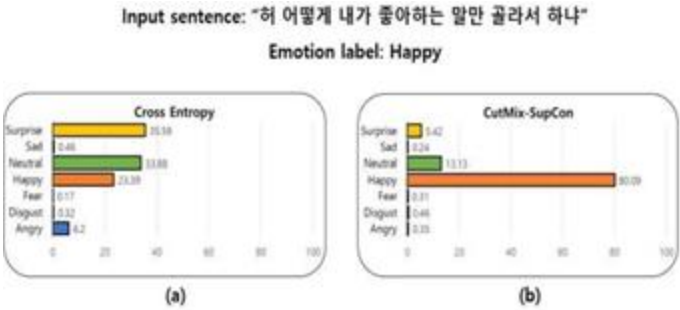


그림 2 KEMDy19 테스트 데이터 샘플에 대한 CE와 CutMix-SupCon의 예측 확률 분포 비교

Adversarial Fine-tuning using Generated Respiratory Sound to Address Class Imbalance

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Introduction

- While *deep generative models* have emerged as a promising approach in the medical image domain to address data scarcity, their use for sequential data like **respiratory sounds** is less explored.
- In this work, we propose a straightforward approach to augment imbalanced respiratory sound data using an **audio diffusion model** as a *conditional neural vocoder*.
- Nevertheless, the **discrepancy between synthetic and real samples** can introduce problems related to *distribution inconsistency*, which can degrade the performance of respiratory sound classification models as the proportion of synthetic data in the training set increases.
- We further introduce a simple yet effective **Adversarial Fine-tuning** method to *align features between the synthetic and real respiratory sound samples* to improve respiratory sound classification performance.

Method 1

- We aim to generate *high-fidelity respiratory sound samples* using audio diffusion model [1] and then **combine these synthetic samples with real data to improve the respiratory sound classification task**, especially for imbalanced lung sound disease classes.
- In phase 1, we introduce a simple method for **generating respiratory sound samples** using a *conditional neural vocoder* inspired by the recent success of *DiffWave* [1] in obtaining realistic audio.

Phase 1: Respiratory Sound Generation

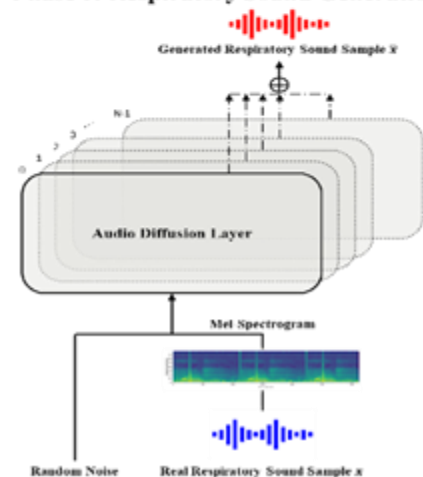


Figure 1: Overview of respiratory sound samples generation using audio diffusion model.

Method 2

- While augmenting real data with synthetic samples can be beneficial, we found in early experiments that in our case the **distribution mismatch between the two types of data degraded the performance of the classification model**.
- To overcome this issue, we propose a simple yet effective **Adversarial Fine-Tuning (AFT)** method inspired by [2].

Phase 2: Adversarial Fine-tuning with Mixed Samples for Respiratory Sound Classification

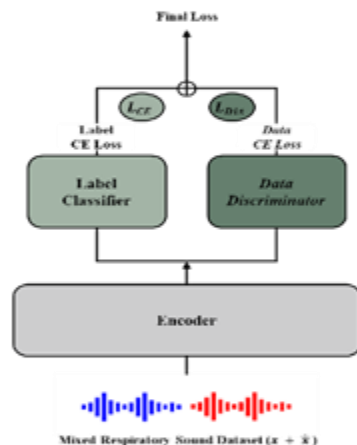


Figure 2: Overview of Adversarial fine-tuning with synthetic and real samples.

- The proposed method consists of two losses with label classifier $\mathcal{L}_{CE}$ and data discriminator $\mathcal{L}_{Dis}$:

$$\mathcal{L}_{CE} = -\frac{1}{N} \sum_{i=1}^N y_i \log(\hat{y}_i), \quad \mathcal{L}_{Dis} = -\frac{1}{N} \sum_{i=1}^N d_i \log(\hat{d}_i),$$

- where $\mathcal{L}_{CE}$ and $\mathcal{L}_{Dis}$ are CE loss with label y and label d , and the predicted probabilities $\hat{y}$ and $\hat{d}$ are obtained by passing through the *classifier* and *discriminator*, respectively.
- To ensure that the *learned features cannot distinguish between the synthetic and real samples*, **gradients from $\mathcal{L}_{Dis}$ are multiplied by a negative constant during the backpropagation**.
- The final training objective is $\mathcal{L}_{Final} = \mathcal{L}_{CE} + \lambda \mathcal{L}_{Dis}$ where λ is a regularization parameter drawn from [2].
- We used Audio Spectrogram Transformer (AST) [3] model as our baseline.

Results

Table 1: Performance according to various mixed sample amounts.

method	training dataset							
	No Aug.	Mixed-500	Mixed-800	Mixed-1k	Mixed-1.5k	Mixed-2k	Mixed-3k	Mixed-5k
AST FT	59.55	59.92 ± 0.82	59.99 ± 1.17	59.81 ± 0.36	59.65 ± 0.30	59.18 ± 0.65	59.04 ± 0.32	58.56 ± 0.84
AFT	-	61.79 ± 0.47	60.89 ± 0.78	60.8 ± 1.05	60.03 ± 1.14	60.64 ± 0.45	59.96 ± 0.38	59.74 ± 0.6

Table 2: Accuracy (%) of the abnormal class on the ICBH test set for AST FT and AFT.

class	ratio	method (dataset)				
		AST FT (No aug.)	AST FT (Mixed-500)	AST FT (Mixed-2k)	Adversarial FT (Mixed-500)	Adversarial FT (Mixed-2k)
crackle (C_c)	23.55%	45.45	42.84	42.86	44.07	46.99
whoeeze (C_w)	13.97%	36.62	36.1	22.08	37.92	30.12
both (C_b)	5.19%	15.38	9.09	7.69	41.96	35.66

Table 3: Overall comparison of the ICBH dataset for the respiratory sound classification task.

method	architecture	pretrain	S_p (%)	S_r (%)	Score (%)
RespireNet [8] (CBA+BRG+FT)	ResNet34	IN	72.30	40.10	56.20
Wang <i>et al.</i> [32] (Splice)	ResNet50	IN	70.40	40.20	55.30
Nguyen <i>et al.</i> [25] (CoTuning)	ResNet50	IN	79.34	37.24	58.29
Moummad <i>et al.</i> [23] (SCL)	CNN6	AS	75.95	39.15	57.55
Bae <i>et al.</i> [3] (Fine-tuning)	AST	IN + AS	77.14	41.97	59.55
Bae <i>et al.</i> [3] (Patch-Mix CL)	AST	IN + AS	81.66	43.07	62.37*
AFT on Mixed-500 [ours]	AST	IN + AS	80.72 ± 0.99	42.86 ± 1.3	61.79 ± 0.47

- While directly fine-tuning on mixed data did not improve the performance of the minority classes overall, **our proposed method improved their accuracies by up to 26.58%, especially in C_b** .
- Furthermore, our proposed method trained with *Mixed-500* achieved a Score of **61.79%**, **outperforming the AST FT model by 2.24%**.

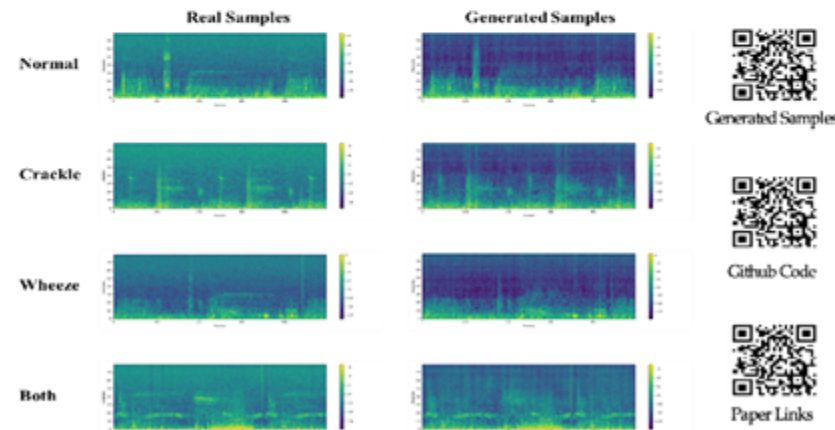


Figure 3: Comparison of spectrograms per each class randomly chosen from the test set and the generated results.

References

- [1] Kong Zhifeng, *et al* (2021). "DiffWave: A versatile Diffusion Model for Audio Synthesis." *ICLR* 2021
- [2] Garin Yaroslav, *et al* (2018). "Domain-Adversarial Training of Neural Networks." *The Journal of Machine Learning Research* 2018
- [3] Gong Yuan, *et al* (2021). "AST: Audio Spectrogram Transformer." *INTERSPEECH* 2023

A Military Audio Dataset for Situational Awareness and Surveillance

Introduction

- Audio classification related to military activities is a challenging task due to the high levels of background noise and the lack of suitable and publicly available datasets.
- To bridge this gap, this paper constructs and introduces a new military audio dataset, named MAD, which is suitable for training and evaluating audio classification systems.
- The proposed MAD dataset is extracted from various military videos and contains 8,075 sound samples from 7 classes corresponding to approximately 12 hours, exhibiting distinctive characteristics not presented in academic datasets typically used for machine learning research.
- We further conduct a comprehensive sound classification study of various deep learning algorithms on the MAD dataset.

Method

- Methodology used to collect the MAD dataset



Fig. 1 Illustration of MAD dataset collection procedure.

- Data selection
Identify the audio content types for the seven military-related categories.
- Audio event segmentation
Identify the start and end times of each audio event.
- Data refinement
Refine the data using a variety of techniques, such as formatting, resampling, and extracting.
- Data labeling
Label all audio events into one of the seven predefined classes.
- Data configuration
Randomly split all the data into train/test sets with an approximately 9:1 ratio.

- Dataset Overview
 - Total Data: 8,075 audio events (12.05 hours in total)
 - Dataset Split

Training Set: 7,393 samples (10.98 hours)

Test Set: 682 samples (1.07 hours)

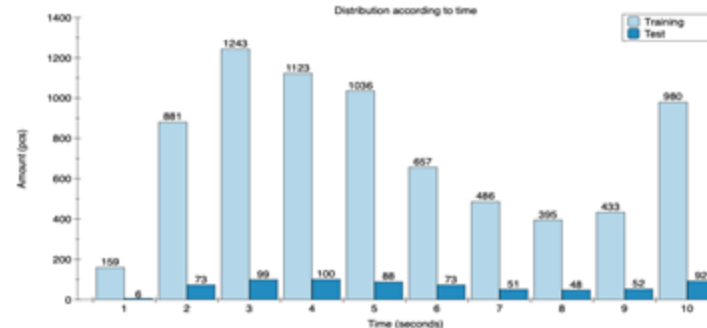


Fig. 3 Bar chart illustrating the distribution of training and test sets according to various times.

- Key Features
 - All audio events range from 1 to 10 seconds in duration. Meticulously labeled and categorized for high-quality deep learning-based audio classification tasks.
 - Total Classes: 7 audio classes, including Communication, Gunshot, Footsteps, Shelling, Vehicle, Helicopter, and Fighter.

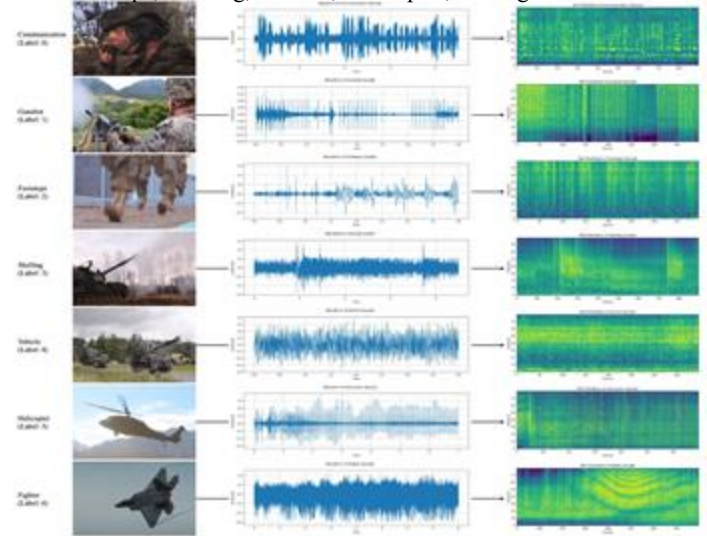


Fig. 4 The overall pre-processing task for converting waveform to Mel filterbank.

Technical Validation

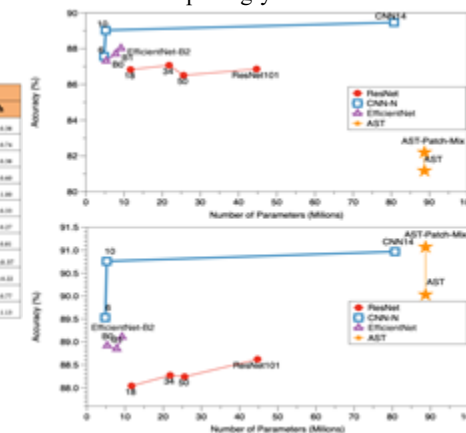
- Model Evaluation: We conducted a comprehensive analysis of various deep learning models to evaluate the performance of the MAD dataset.
 - Models: ResNet, EfficientNet, CNN-N (PANNs), and AST (Audio Spectrogram Transformer).
- Pre-processing
 - Audio waveforms were converted into Log Mel filterbank features (80 dimensions, 25ms window, 10ms overlap).
 - All audio samples were standardized to a 10-second length using zero padding.
 - Standard normalization was applied to the spectrograms.
 - Augmentation: SpecAugment was used to mitigate overfitting.



Fig. 5 Illustration of overall deep learning-based audio classification process.

- Experimental Result Summary
 - Pretrained models consistently outperformed those trained from scratch. With pretrained weights, the AST-Patch-Mix model achieved the highest accuracy (91.07%), aligning with the trend that models with more parameters perform better. A notable exception was the CNN10 model, which achieved a strong 89.53% accuracy with only 6% of the parameters of the AST model.
 - In contrast, when trained from scratch, the CNN14 model performed best (89.47%), while the high-parameter AST models surprisingly showed the lowest performance.

architecture	parameters	pretrain	accuracy (%)	pretrained weights	scratch
ResNet18	11.7M		88.04 _{±0.08}		86.83 _{±0.08}
ResNet34	21.8M	ImageNet	88.27 _{±0.08}		87.07 _{±0.08}
ResNet50	25.4M		88.24 _{±0.08}		86.51 _{±0.08}
ResNet101	44.7M		88.62 _{±0.08}		86.86 _{±0.08}
EfficientNet-B0	5.3M		88.92 _{±0.07}		87.32 _{±0.08}
EfficientNet-B1	7.8M	ImageNet	88.85 _{±0.08}		87.71 _{±0.08}
EfficientNet-B2	9.2M		89.51 _{±0.08}		88.02 _{±0.07}
CNN6	4.8M		89.53 _{±0.08}		87.57 _{±0.08}
CNN10	5.2M	AudioSet	90.76 _{±0.07}		89.03 _{±0.07}
CNN14	86.7M		90.37 _{±0.08}		89.47 _{±0.08}
AST	87.7M	ImageNet + AudioSet	90.03 _{±0.08}		81.17 _{±0.07}
AST-Patch-Mix		ImageNet + AudioSet	91.07 _{±0.08}		82.20 _{±0.08}



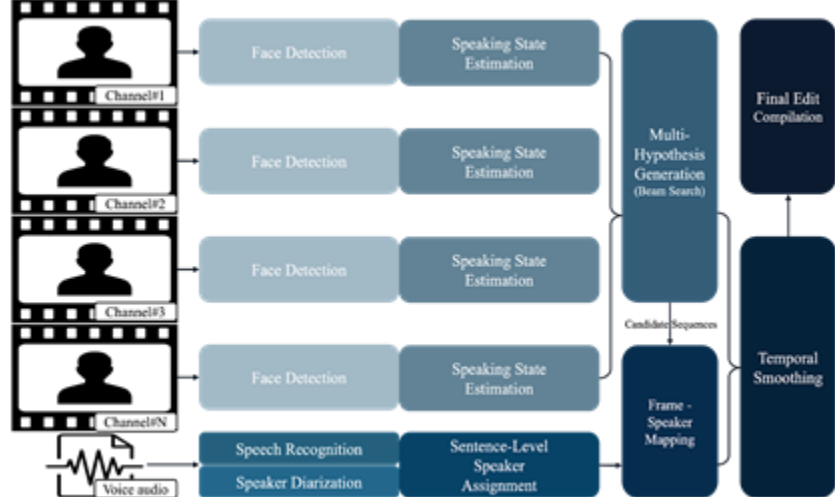
Seamless Multi-Hypothesis Video Editing Using Deep Multimodal Analysis Over Multi-Channel Sources

Introduction

- An automated system for editing multi-channel video by synchronizing raw inputs through separate visual and auditory pipelines.
- Deep late-fusion of multimodal information for face recognition, speaking-state estimation, speech recognition and speaker identification.
- A multi-hypothesis editing approach that provides candidate versions based on probability distribution over time for extracting frames from each channel stream.
- Frame-level match accuracies of 84.99% and 90.10% for the 3-channel and 5 channel videos, highlighting the system’s robust performance across varying speaker complexities.

Method

- For multi-speaker, multi-channel content, we adopted an intuitive approach to make automated editing feasible. This is primarily because datasets containing multi-speaker, multi-channel videos paired with their fully edited versions are extremely rare.
- The lack of such datasets makes it challenging to apply standard supervised learning techniques effectively for automated video editing.
- For the audio modality, our goal is to accurately detect active speech segments and identify the corresponding speakers from diverse and often noisy multi-channel sources. - Speech Recognition, Speaker Diarization
- The visual approach involves estimating the speaking state based on the facial region of the speaker. - Face Detection, Speaking State Estimation
- Integration and Channel Selection - Multi-Hypothesis Generation for Temporal Coherence



Experiment & Result

- Dataset
 - Studio recordings provided by a Korean public broadcasting company in South Korea. {3-channel video dataset, 5-channel video dataset}
- Evaluation
 - Compare the ground-truth labeled data with output generated by our system at the frame level.
 - For the Quantitative Analysis, Analyze the frame-level alignment accuracy under various smoothing and temporal conditions.

Data	Thr 15	Thr 30	Thr 45
Best Path (self-transition bias 0.7)			
3-channel video	0.8498	0.8495	0.8495
5-channel video	0.8995	0.8999	0.8854
Best Path (self-transition bias 0.9)			
3-channel video	0.8499	0.8497	0.8481
5-channel video	0.8987	0.9010	0.8865

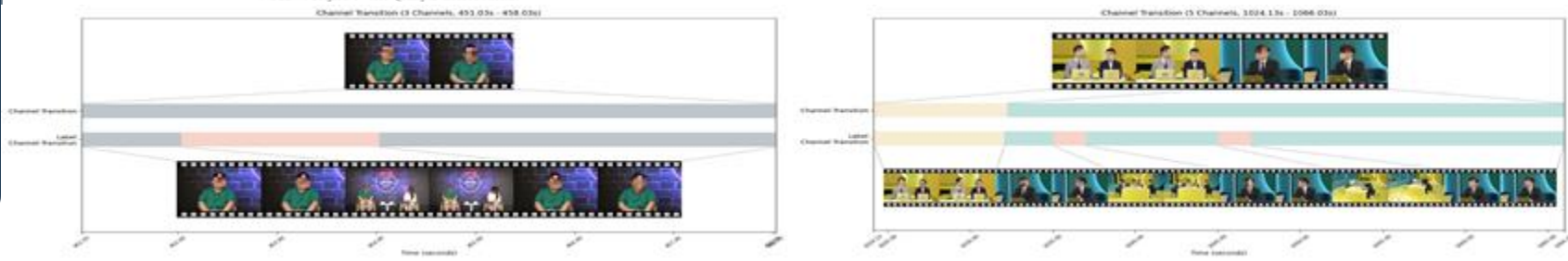
TABLE I: Match rates (Top Beam Paths applying self-transition bias 0.7/0.9) varying the smooth threshold.

Length (s)	3-channel video			5-channel video		
	Match	Frames	Rate	Match	Frames	Rate
0-5	3,849	6,660	0.5779	2,981	4,335	0.6877
5-10	4,162	4,695	0.8865	5,447	5,940	0.9170
10-20	5,161	5,340	0.9665	2,867	4,425	0.6479
20-30	2,971	3,210	0.9255	3,573	3,720	0.9625
30-40	1,140	1,140	1.0000	8,412	8,580	0.9804
40-50	—	—	0.0000	2,524	2,550	0.9898
50-60	3,104	3,315	0.9363	1,648	1,650	0.9988
60-70	—	—	0.0000	2,010	2,010	1.0000
70-80	2,128	2,130	0.9991	2,198	2,220	0.9901
80-90	—	—	0.0000	2,670	2,670	1.0000
Total	22,515	26,490	0.8499	34,330	38,100	0.9010

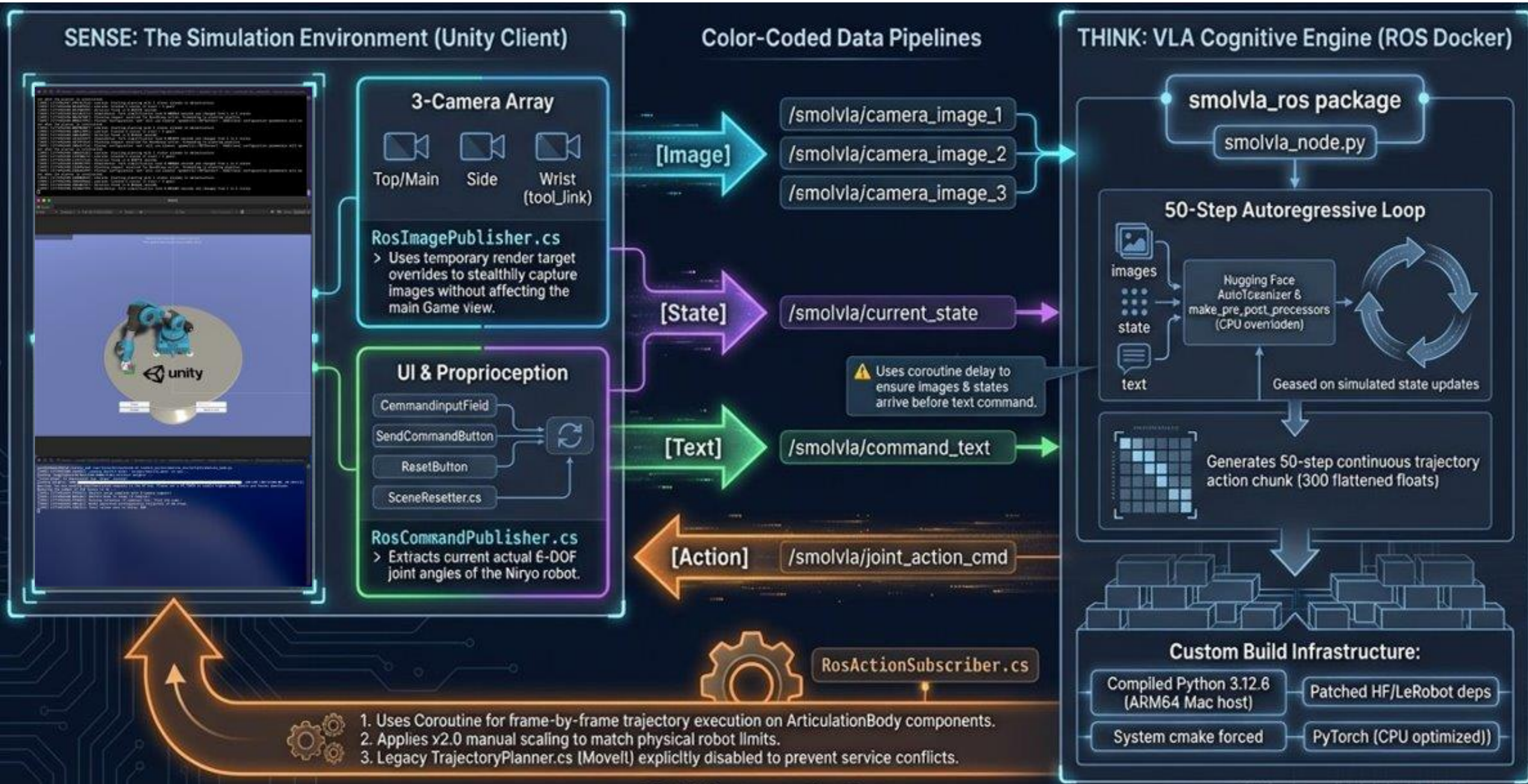
TABLE II: Match rates across different segment lengths.

Criteria	3-channel		5-channel
	Ours	Expert	Expert
1. Speaker Switching Smoothness & Seamlessness			
1.1 Is the camera transition smooth when the speaker changes in the video?	4.26±0.81	4.43±0.78	3.95±1.02
1.2 Are the cut transitions in the video not awkward?	4.26±1.00	4.34±0.64	4.04±1.02
2. Scene Consistency			
2.1 Are the scenes in the video consistently focused on the speaker in context?	4.17±0.88	4.52±0.59	3.95±0.92
3. AI Editing Potential			
3.1 If the AI-edited video can replace professional editing videos?	—	—	4.08±1.08
3.2 If the AI automatic editing system can help professional editors?	—	—	4.56±0.66

TABLE III: Subjective Video Quality Evaluation Results.



Unity Robot Integration with SmolVLA

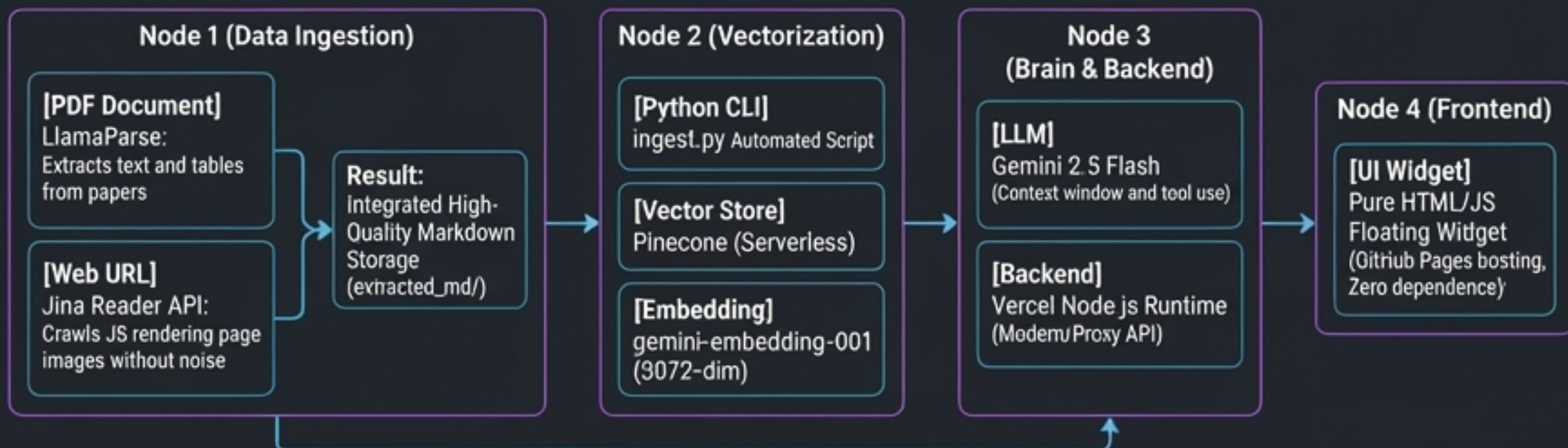


Chihyeon.ai - Personal RAG Assistant

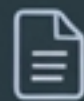
A high-performance Serverless AI assistant operating based on AI Researcher Chihyeon Yun's knowledge base.

A lightweight, reactive RAG chatbot system that autonomously parses and embeds personal papers, portfolios, and web content, providing accurate answers based on Gemini 2.5 Flash.

System Architecture Pipeline



3 Key Pillars



High-Fidelity Document Parsing

Extracts complex layouts, tables, formulas, and images without noise, maximizing RAG retrieval accuracy with clean Markdown.



Lightweight Drop-in UI

A lightweight Vanilla JS-based widget that doesn't conflict with existing frontend CSS. Smooth, real-time streaming via Fetch API and SSE (Server-Sent Events).



Secure & Serverless Architecture

Utilizes Vercel proxy and backend-only API Key storage for zero server maintenance costs. Zero latency management via dialogue history.

Engineering Troubleshooting



Problem 1: CORS and access blocking in certain regions (China/Hong Kong) (Cloudflare Workers limitation)



Solution: Migrated to Vercel (US base) infrastructure and implemented Pure JS to completely overcome regional blocking and ESM dependencies.



Problem 2: Previous embedding model deprecation and decline in search accuracy.



Solution: Quickly migrated from text-embedding-004 to high-dimensional gemini-embedding-001 (3072-dim) to enhance search performance.

Tech Stack

AI Models: Gemini 2.5 Flash Gemini Embeddings

Data & DB: Pinecone LlamaParse Jina Reader

Backend: Vercel Node.js Python

Frontend: Vanilla JS HTML/CSS3 SSE